

ELECTIVE COURSE-9.3

UCS519: SOFTWARE DESIGN PRINCIPLES				
	L	T	P	Cr
	2	0	2	3.0
Course Objective:				
Syllabus Software Models and Design Principles: The Role of Models in Engineering, Characteristics of Good Engineering Models, Models of Software, UML Profiles, System design principles like levels of abstraction, separation of concerns, information hiding, coupling and cohesion, Structured design (top-down or functional decomposition), object-oriented design, event driven design, component-level design, test driven design, data design at various levels, Structural Modelling with UML, Classes, Domain Model, Behaviour Model for Interaction, Modelling with Internal Structure, Decomposition, Consistency of Interactions and State Machines Message Sequence Charts (MSCs): Basic MSCs, High Level MSCs, Message Sequence Charts, scenarios, Live Sequence Charts, Control Constructs, THE PLAY-IN/PLAY-OUT APPROACH, The Application Section, The Specification Section, The GUI Application Communicating Transaction Processes (CTP), Control Flow of CTP, Transaction Scheme of CTP, executable specification Hardware and Software Object Modeling for Embedded System Development: System Development, Methods, The HASOC Design Lifecycle, SystemC, object-orientation, lifecycle modeling, platform modelling, Software Hardware Interface Model, Hardware Architectural Model, (Meta) Model-Based Development Processes Platform-based Design: platform-based design, embedded system design, The Metropolis design environment. Modelling Platforms using UML, Stereotypes, UML Platform Design Methodology, Design Problem Formulation, Use Case Diagram. Functional Specification, Key element of Platform Model, Communication Requirement Real-Time Systems Design: Qualitative Real Time Features, Concurrency Modelling, Active Objects, Process, Threads, operations, communication modelling, Operation-based messaging, Signal-based messaging, Behavioural Modelling, State Chart Diagram, Activity Diagrams, Sequence Diagram Automotive UML: Reconciling the Needs of Automotive Software Development with Model-Based Approaches, Abstraction Levels, Architecture Description Language (ADL), Automotive Modelling Language (AML), Electronic Control Unit (ECU), Metamodel, Requirements Engineering, AML for Automotive Systems Development, Function, Ports and Connectors, Role of AML Meta-Model, UML Notations for AML, Functional and Non-Functional Requirements, Functional Network.				

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively. Further, revision Approved in 1st meeting of Academic Council held on September 11, 2025.

Specifying Telecommunications Systems with UML: Telecommunications, ITU, Service Description, Service Description Methodology, System Design Language (SDL) for Telecom profiles, Leveraging UML to Deliver Correct Telecom Applications, Verification and Validation

Deployment: Component, Dependencies, Packages, Component Diagrams, Nodes, Associations, Deployment Diagrams, Patterns and frameworks.

Course Learning Outcomes (CLOs)

The students will be able to:

1. recognize principles and fundamentals of software design to identify, analyse, visualize, specify, model and design
2. apply analysis and design principles at various levels and various views in different domains of software systems.
3. represent and generate design models for real time applications, embedded systems, automotive, and telecommunications.
4. skillful use of rational rose tool for drawing all the UML diagrams in order to forward and reverse engineer the complex engineering problems.

Text Books

1. “UML for Real: Design of Embedded Real-Time Systems” Bran Selic, Luciano Lavagno and Grant Martin , Kluwer Academic Publishers 2nd Ed. (2003)
2. “Real Time UML: Advances in the UML for Real-Time Systems” Bruce Powel Douglas, Pearson 3rd Ed., (2004)
3. “The Unified Modeling Language User Guide” Grady Booch, James Rumbaugh, and Ivar Jacobson, Addison-Wesley Professional, 2nd Ed (2005).

Evaluation scheme

Sr. No.	Evaluation Elements	Weightage (%)
1	MST	25
2	EST	40
3	Sessional (May include Assignments//Quizzes/Lab Evaluations)	35

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